

Using QGIS

Week 4 — Tuesday Lecture

Today

- What a GIS is, and why an orthomosaic is a measuring instrument
- Installing QGIS — before Thursday
- Loading an orthomosaic and checking the coordinate reference system
- Two ways to measure: the quick one, and the one you can hand in

A picture, or a measurement?

An aerial photo shows you a parking lot. An orthomosaic in a GIS tells you it is **27,856 ft²**.

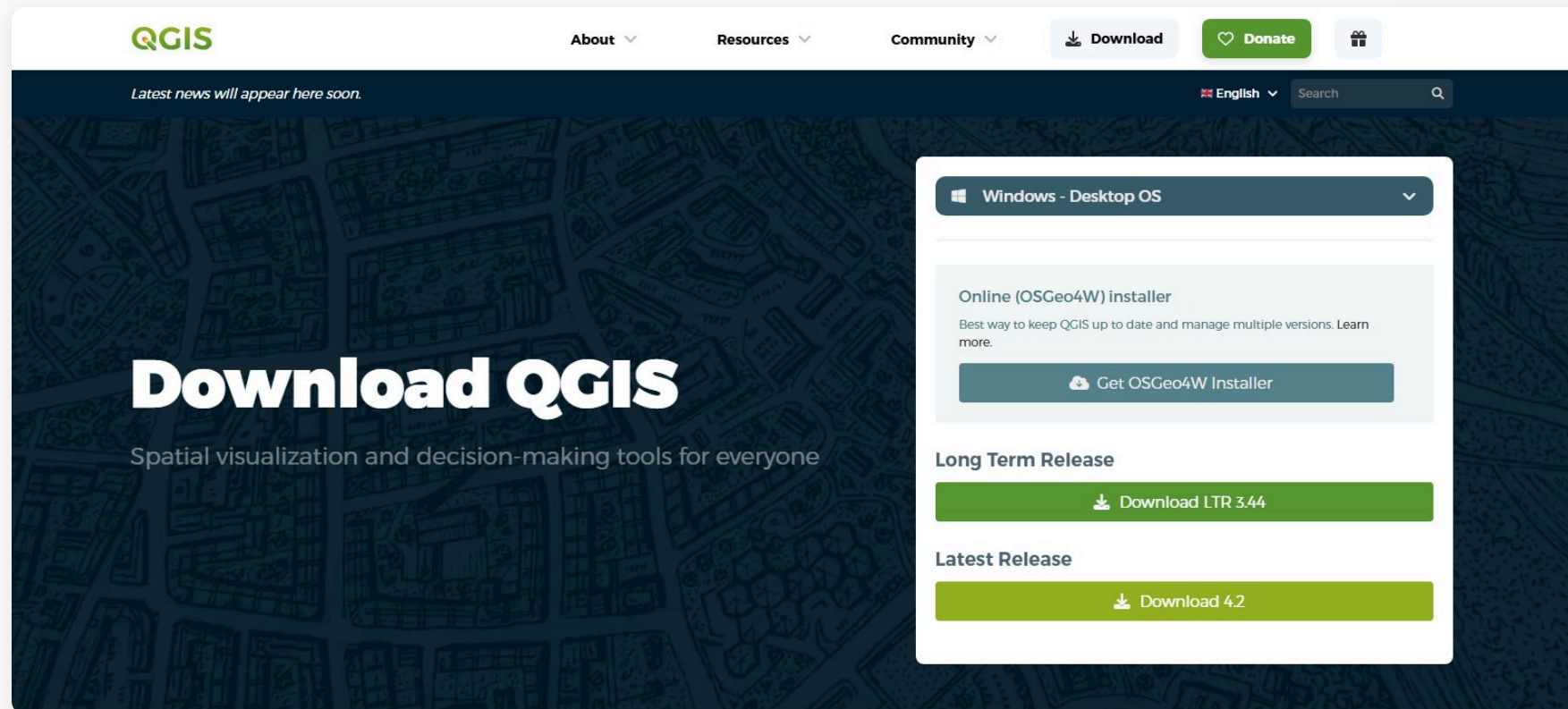
What has to be true of the image for that number to mean anything?

GIS is a database with a map on the front

- Every pixel of an **orthomosaic** has a real-world coordinate
- So a click on the screen is a **position**, two clicks are a **distance**, and a closed shape is an **area**
- **QGIS** is the free, open-source GIS we use to take those measurements

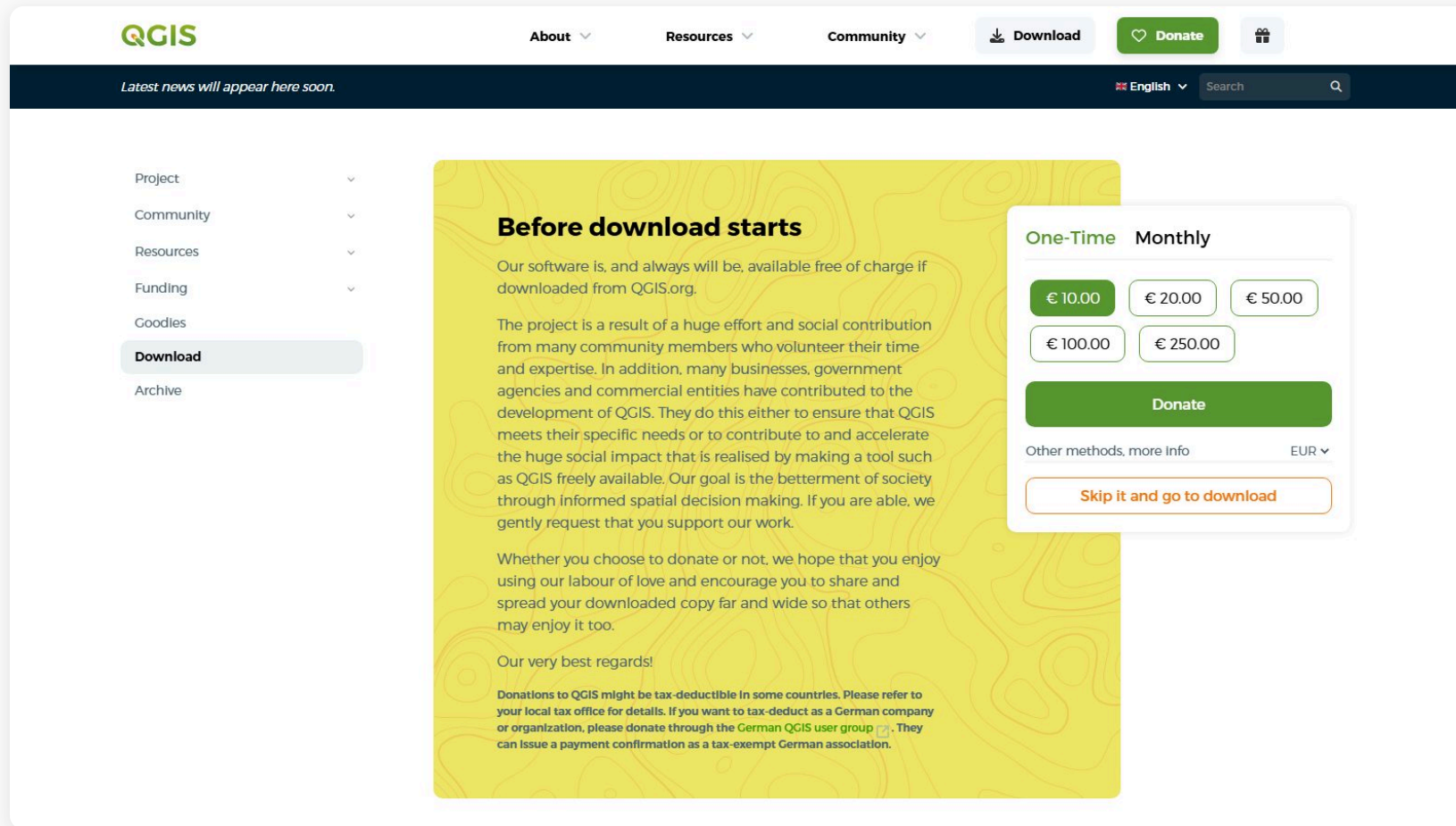
The number is only as good as the image under it and the coordinate system it is drawn in.

Install QGIS



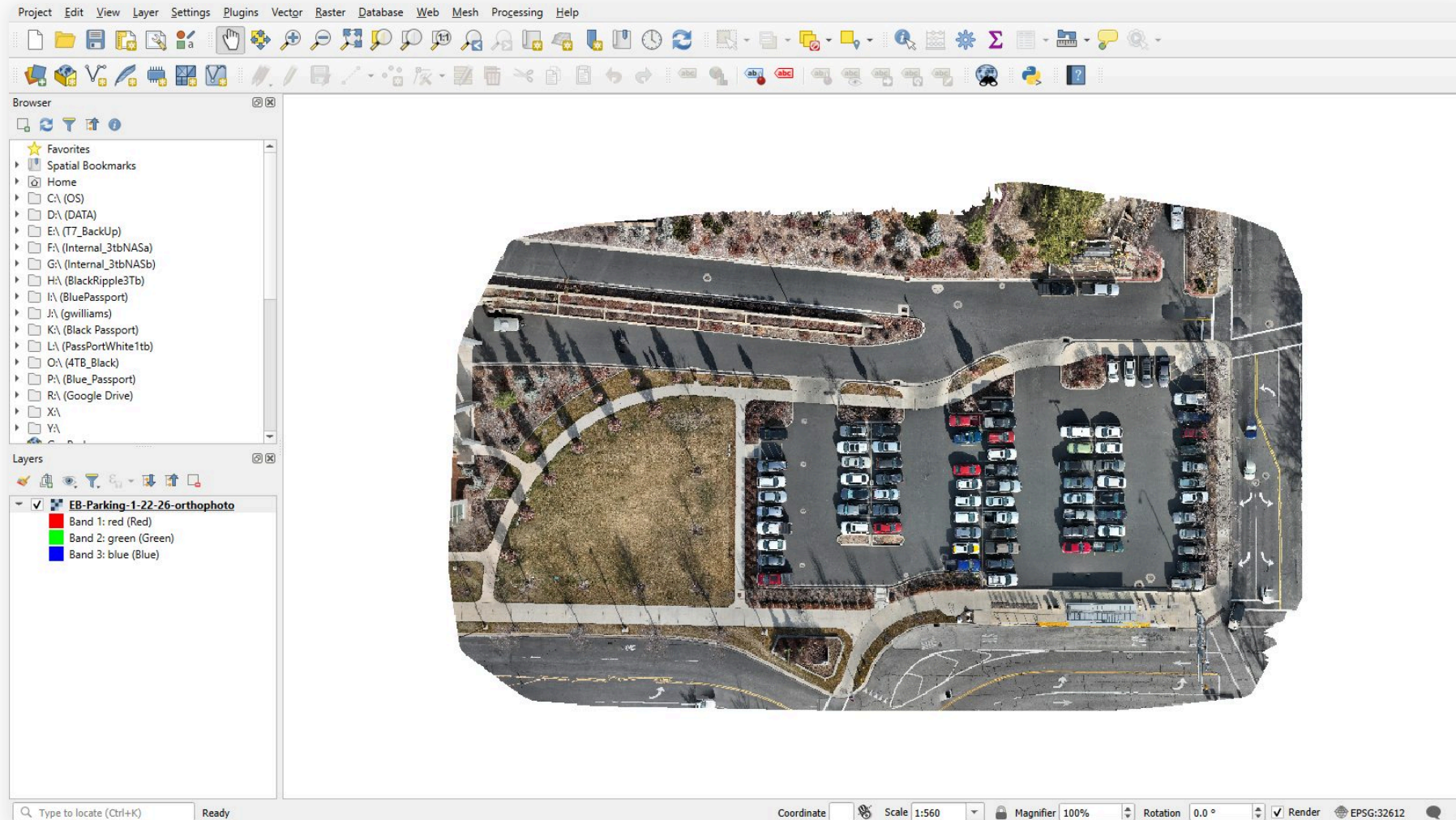
qgis.org → Download → **Long Term Release (LTR)**. Windows: run the .msi and keep the defaults.

It will ask for a donation first



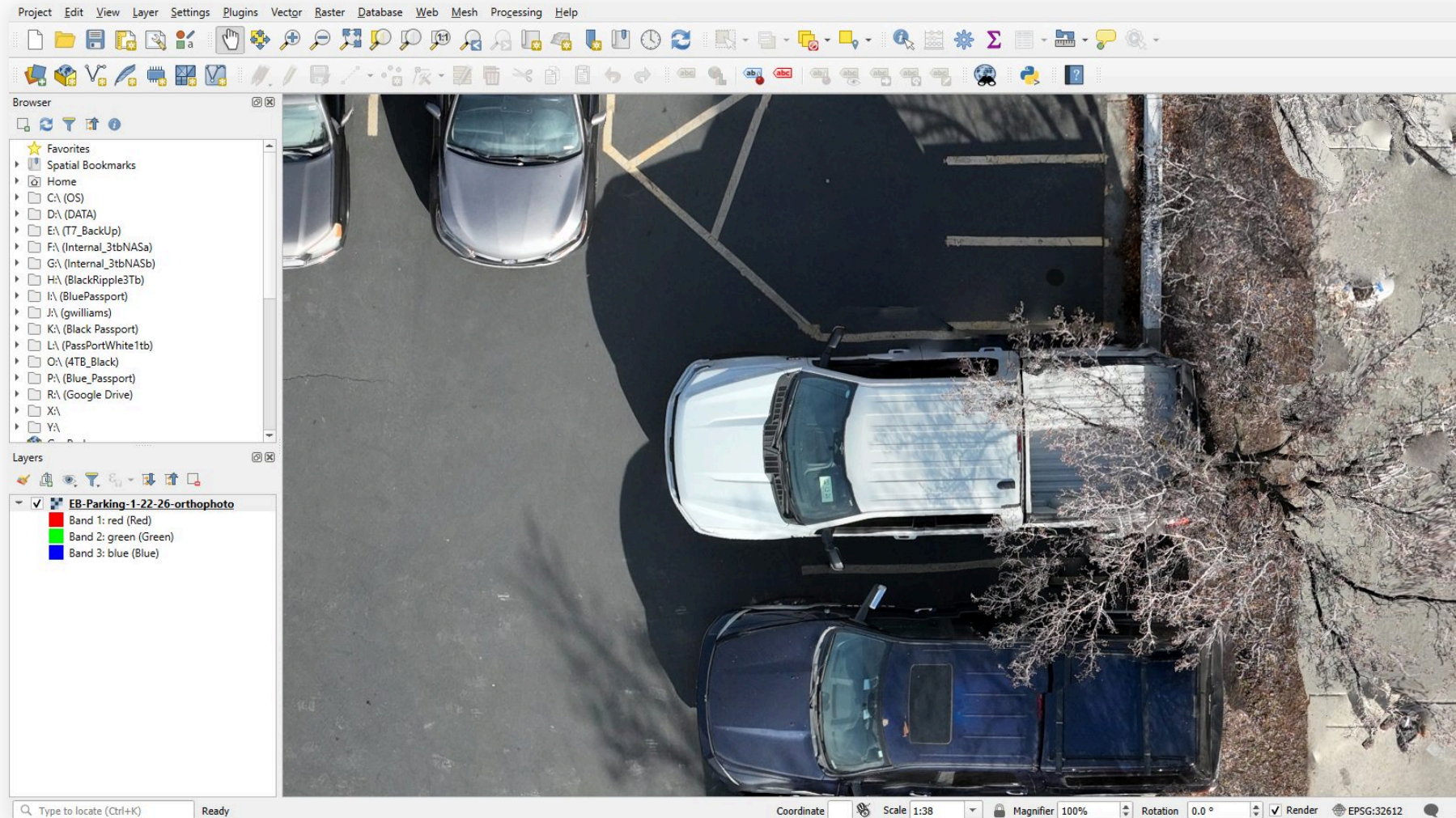
Not required. Click **Skip it and go to download.**

Load the orthomosaic



New empty project, then drag the .tif onto the map. Layers panel on the left; the CRS in the bottom-right corner.

Zoom in

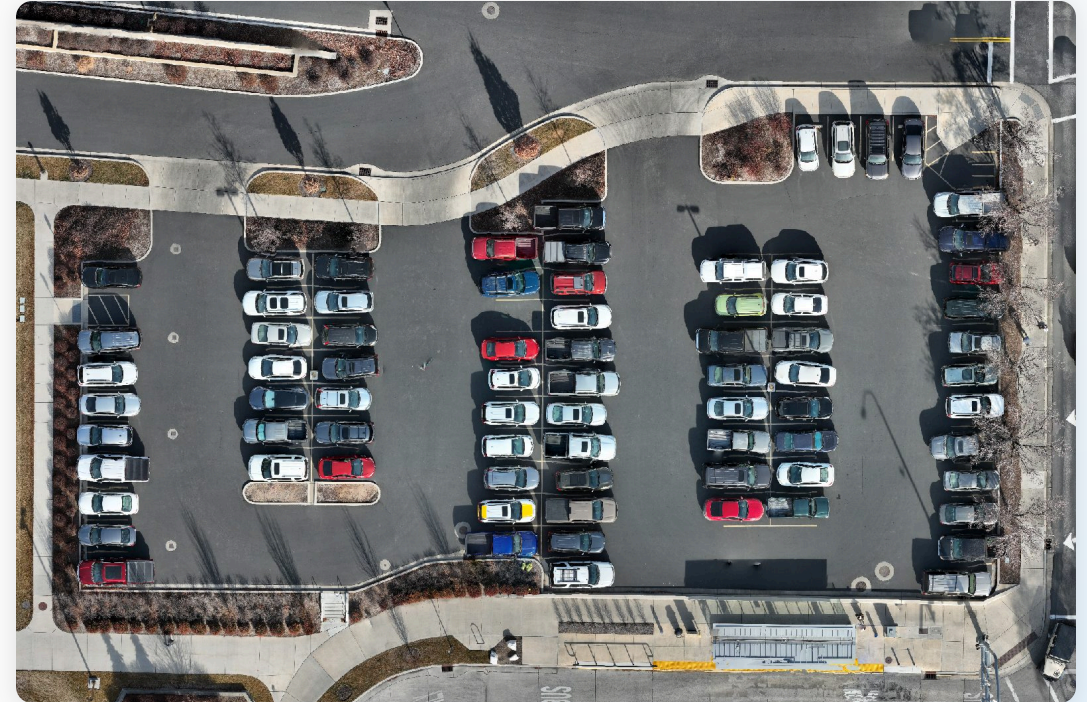


One-centimetre pixels. You can see the stripe, the curb, and the bumper — and that matters for where you click.

Same lot, two images



Google Maps satellite imagery (© Google)



Drone orthomosaic, January 2026, 1 cm pixels

Where does the pavement end?

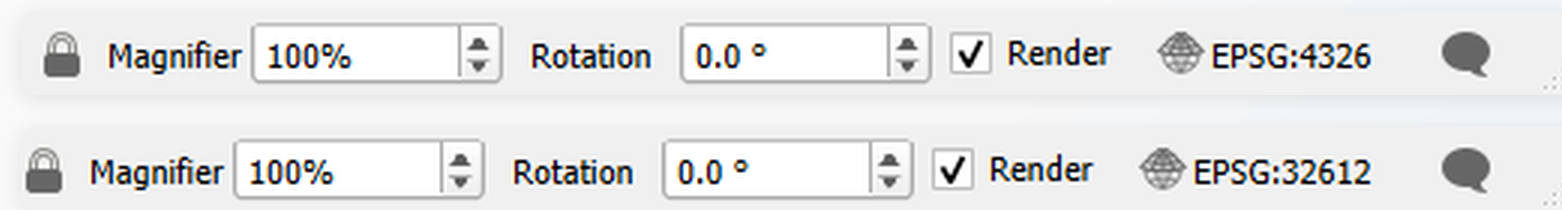
Zoom in on both. Which one lets you put the cursor on the curb with confidence — and how much area is one metre of doubt around the whole lot?

The coordinate reference system

- A **CRS** is how the round Earth is flattened onto your screen
- **Geographic** CRS (plain WGS 84, `EPSG:4326`): units are **degrees**
- **Projected** CRS (WGS 84 / UTM zone 12N, `EPSG:32612`): units are **metres**

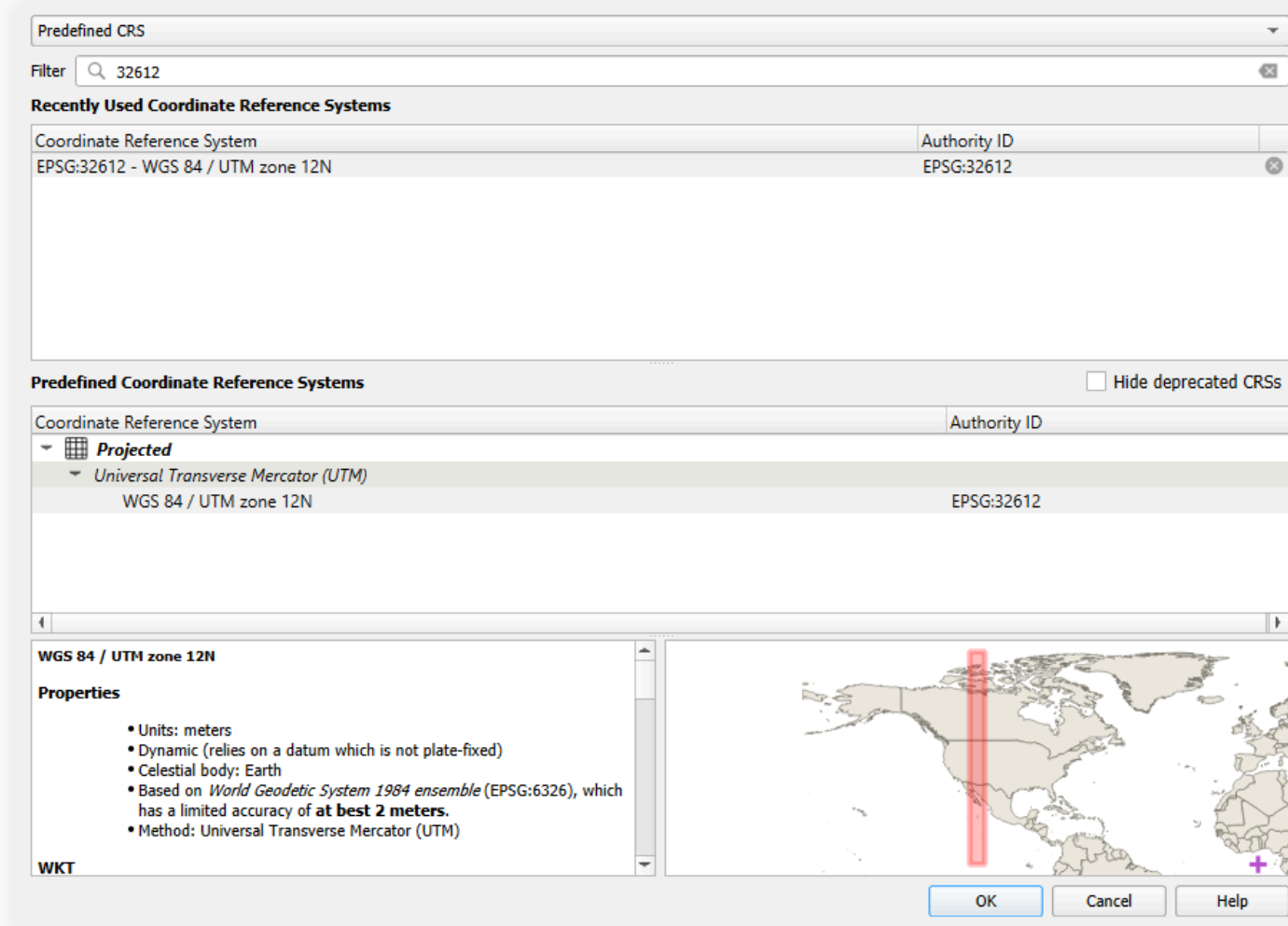
Measure in degrees and every number you write down is meaningless. Check the corner **before** you measure.

The corner tells you



Same project, two settings. Top: degrees — do not measure. Bottom: metres — go ahead.

Set the project CRS



Click the corner (or Project → Properties → CRS), type **32612**, pick WGS 84 / UTM zone 12N.

Set the units you will report in

The screenshot shows the 'General Settings' and 'Measurements' panels of a software application. The left sidebar contains icons for General, Metadata, View Settings, CRS, Transformations, Styles, Colors, Data Sources, Relations, Variables, and Macros. The 'General Settings' panel includes fields for Project file, Project home, Project title, Selection color (set to yellow), Background color, Save paths (set to Relative), and two checkboxes: 'Avoid artifacts when project is rendered as map tiles (degrades performance)' and 'Remember attribute tables windows and docks between sessions'. The 'Measurements' panel includes a dropdown for Ellipsoid (set to WGS 84 (EPSG:7030)), input fields for Semi-major (6378137.000) and Semi-minor (6356752.314), a dropdown for Units for distance measurement (set to Feet), a dropdown for Units for area measurement (set to Square Feet), and a dropdown for Scale calculation method (set to Calculate along Middle of Map).

General Settings

Project file

Project home

Project title

Selection color Background color

Save paths

☐ Avoid artifacts when project is rendered as map tiles (degrades performance)

☐ Remember attribute tables windows and docks between sessions

Measurements

Ellipsoid

Semi-major Semi-minor

Units for distance measurement

Units for area measurement

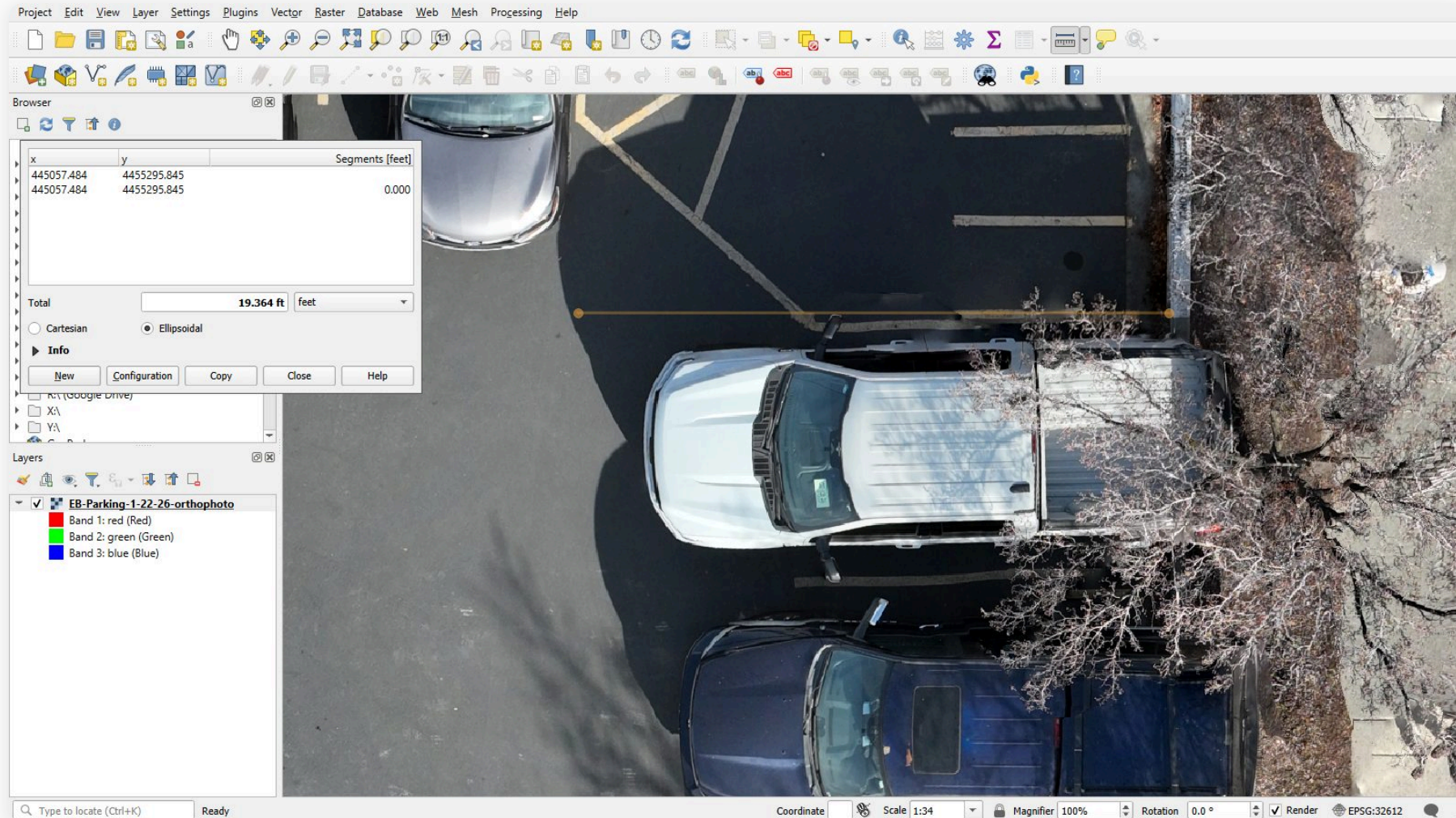
Scale calculation method

Project → Properties → General → Measurements: **feet** and **square feet** for the lab worksheet.

A parking lot of 0.00002

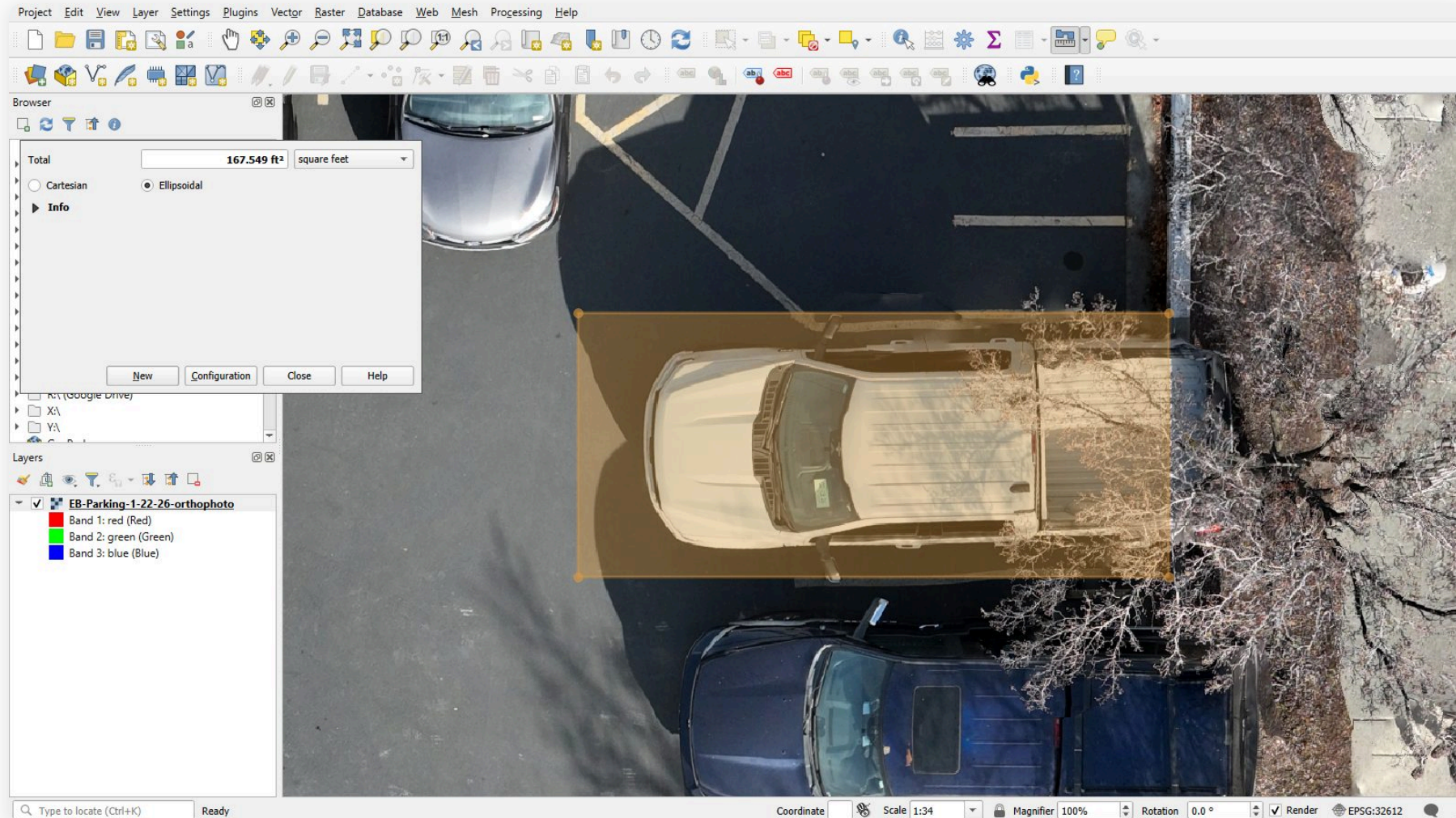
That is what a student's area comes out as, every semester. What happened, and what is the unit?

Measure a distance



Ruler icon or Ctrl+Shift+M. Click, click, **right-click** to finish. One stall: 19.4 ft.

Measure an area



Measure Area from the ruler's drop-down. Click each corner, right-click to close. One stall: 167.5 ft².

The ruler challenge

Measure the stall in **metres**. Switch the units to **feet** and measure it again.

Did the number change? Did the stall?

The Measure tool forgets

The number is on screen, and then it is gone.

For anything you have to **defend** — a lab result, a quantity in a report — draw the shape instead:

1. Create a polygon layer
2. Trace the feature
3. Let the Field Calculator compute the area

The shape is saved with the project. Reopen it, fix one edge, hand it in.

Create a polygon layer

File name: C:\Users\student\Documents\lab2\parking_lot.shp

File encoding: System

Geometry type: Polygon

Additional dimensions: ☒ None ☐ Z (+ M values) ☐ M values

CRS: EPSG:32612 - WGS 84 / UTM zone 12N

New Field

Name:

Type: abc Text (string)

Length: 80 Precision:

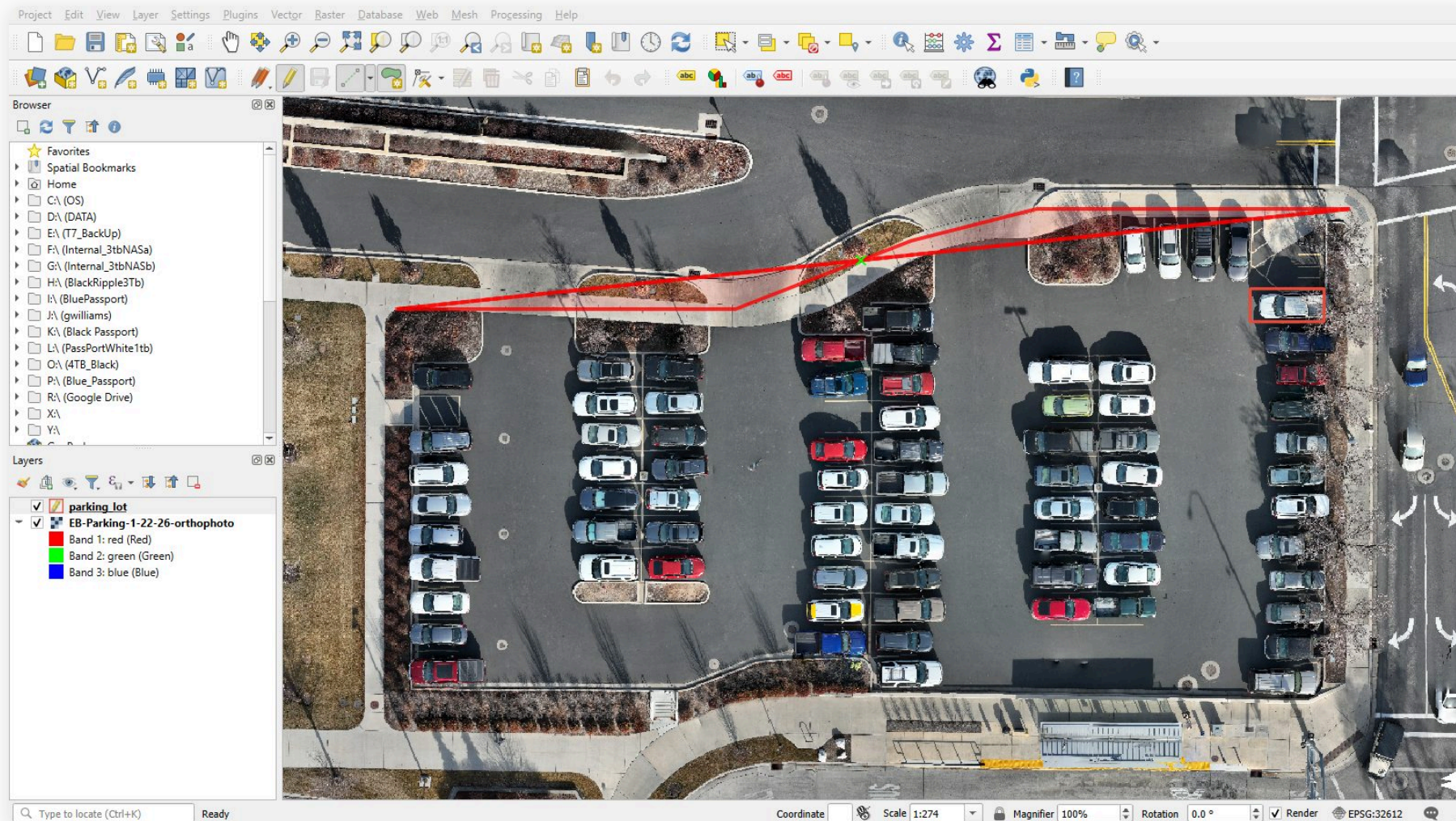
Add to Fields List

Fields List

Name	Type	Length	Precision
id	Integer	10	

Layer → Create Layer → New Shapefile Layer. Geometry **Polygon**, CRS **EPSG:32612**, a file name you can find again.

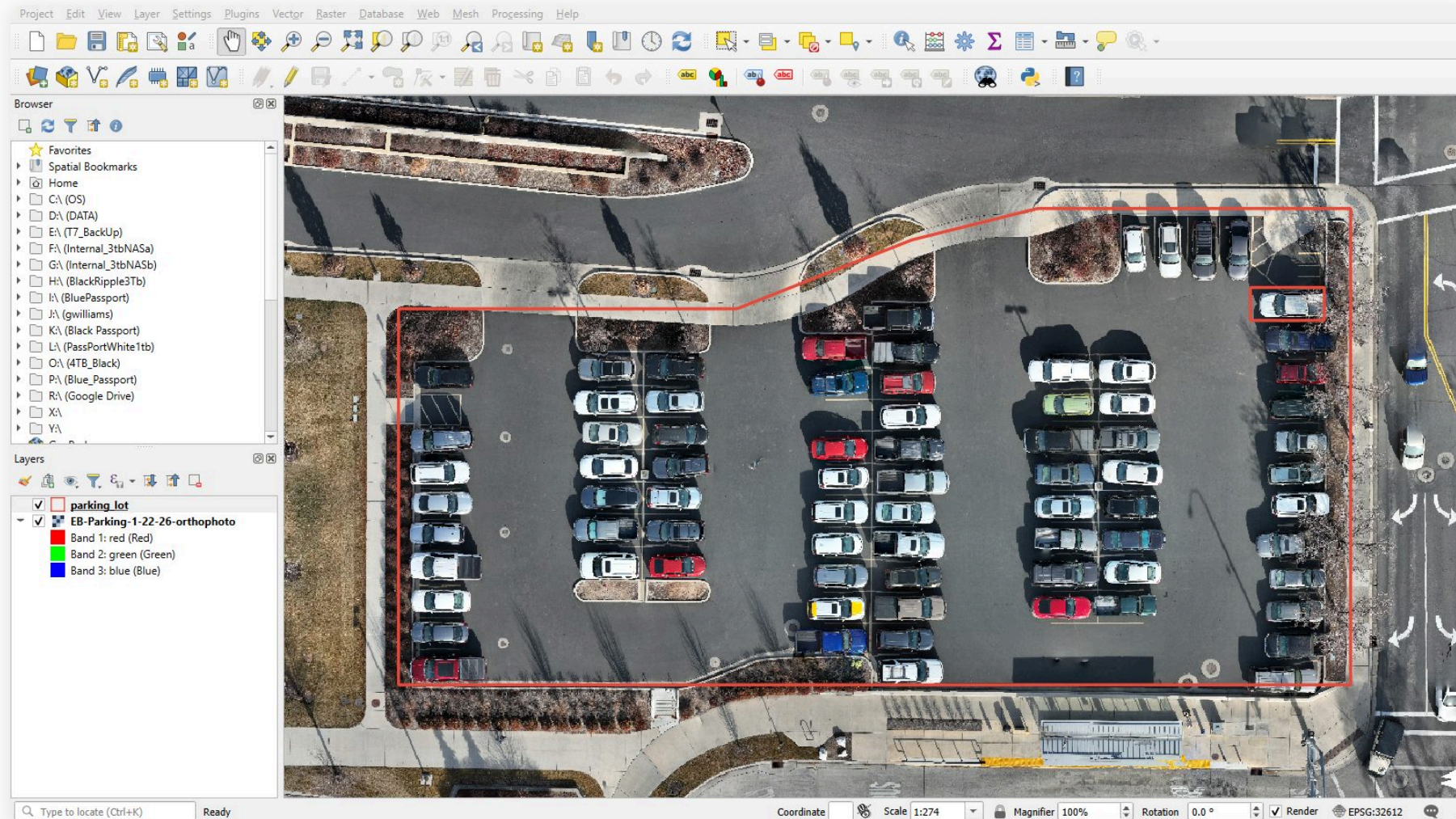
Trace the feature



Toggle Editing (pencil), Add Polygon Feature, click along the asphalt edge, right-click to close. Asphalt only — no sidewalk, no grass.

CCE 194R — Aerial Measurement

Save it



Toggle Editing off and save. Two shapes on record: the stall and the lot.

Compute the area

☐ Only update 0 selected features

☒ **Create a new field** ☐ **Update existing field**

☐ Create virtual field

Output field name:

Output field type:

Output field length: Precision:

Expression:

Function Editor:

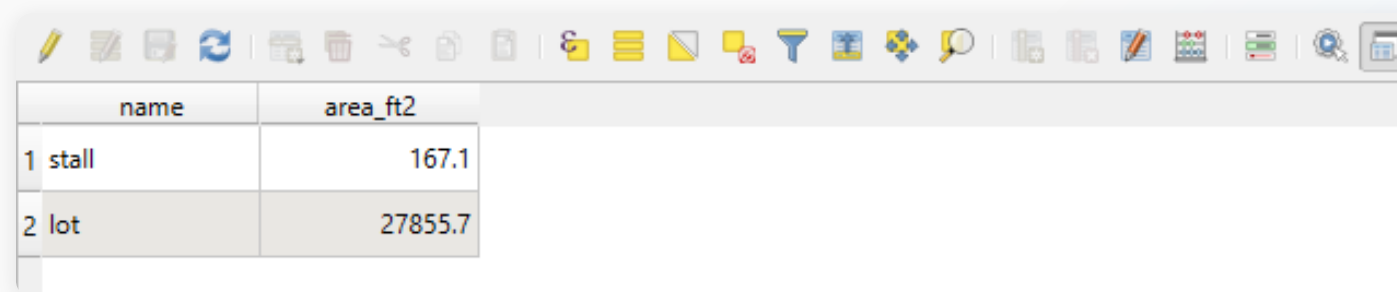
Feature: Preview: 27855.680767793037

Warning: You are editing information on this layer but the layer is currently not in edit mode. If you click OK, edit mode will automatically be turned on.

OK Cancel Apply Help

Open the Attribute Table, then the Field Calculator (abacus). New field **area_ft2**, decimal, expression **\$area**.

Read it off



name	area_ft2
1 stall	167.1
2 lot	27855.7

Stall 167.1 ft², lot 27,856 ft². These are the numbers that go on the worksheet.

Two methods, one feature

Method	Stall area
Measure tool	167.5 ft ²
Digitized polygon + \$area	167.1 ft ²

They agree to a quarter of a percent. If yours differ by a factor of thousands, **check the CRS first** — nothing else produces a mistake that large.

Thursday

- **Install QGIS before lab** — the download is large and the lab Wi-Fi is not
- Bring a laptop; the orthomosaic download is on the lab page
- You will pace the lot, measure it in Google Maps, then measure it in QGIS — and compare all three

Measurements and Methods · Using QGIS